



Mathematics 2 (5EZB0)

Studio classroom 1 – Week 2

April 30, 2026

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Studio classroom

- Recap some of the topics covered in the lectures
- Cover **relevant new bits and pieces of theory**
- Relate mathematics concepts to engineering examples (Physics for EE/AT, Electromagnetics I)
- Work together on suitable examples
- Active participation: some time to try yourselves, then plenary solution on blackboard

Goal: show that math is relevant and fun at the same time!

Studio classroom (cont'd)

Incentive for attending the sessions:

- Learn how to put math to good/relevant use
- Problems in tests on application of the mathematics

Topics for today

- Coordinate systems and their unit vectors
(Adams 10.5 + extra)
- Vector functions of one variable and parametrizations
(Adams 12.1 & 12.3)

Coordinate systems

Cartesian, cylindrical, spherical
...and their unit vectors

Coordinate systems

Properties of coordinate systems:

- unique representation of a point in 3D-space
- directions orthogonal in each point
- convention: choose right-handed

Commonly used coordinate systems for engineering:

- Cartesian or rectangular coordinates
- circular cylindrical coordinates
- spherical coordinates

Cartesian coordinates

Unit vectors:

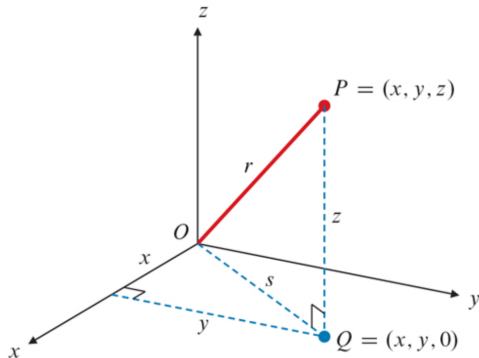
$$\hat{i}, \hat{j}, \hat{k}$$

Position vector:

$$\mathbf{r}_P = x\hat{i} + y\hat{j} + z\hat{k}$$

Vector field:

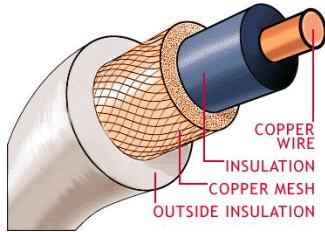
$$\mathbf{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$



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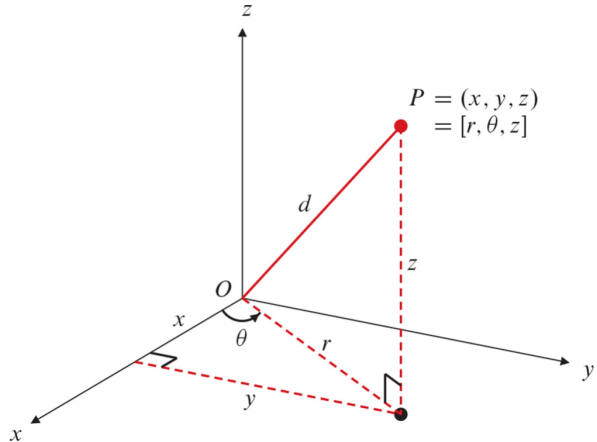
Note: unit vectors relate to small changes in position vector with x , y , and z coordinates

Circular cylindrical coordinates



Represented by $[r, \theta, z]$

$$r \geq 0, \quad \theta \in [-\pi, \pi], \quad z \in \mathbb{R}$$



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Circular cylindrical coordinates

Definition:

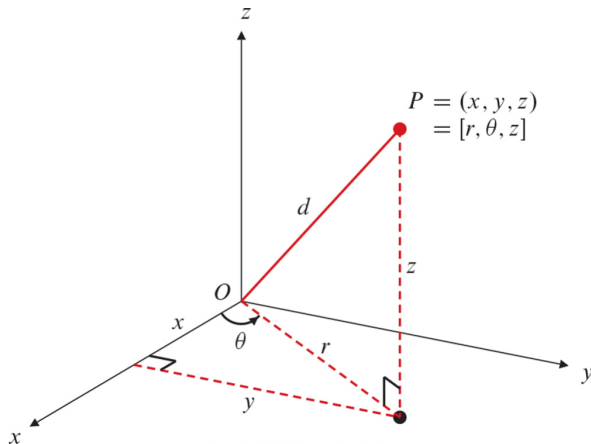
$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$



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Circular cylindrical coordinates – vectors

Unit vectors:

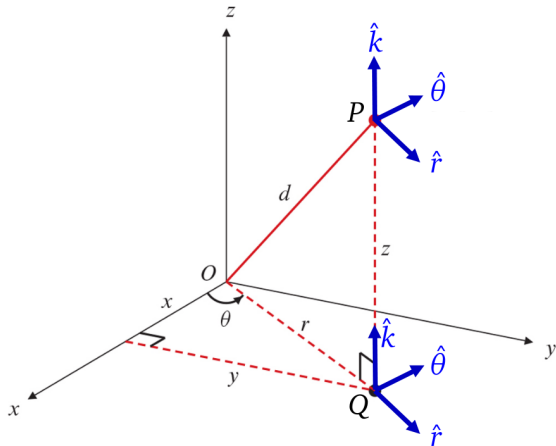
$$\hat{r}, \hat{\theta}, \hat{k}$$

Position vector:

$$\mathbf{r}_P = r \hat{r}(\theta) + z \hat{k}$$

Vector field:

$$\mathbf{A} = A_r \hat{r} + A_\theta \hat{\theta} + A_z \hat{k}$$



Note: unit vectors relate to small changes in position vector with r , θ , and z coordinates

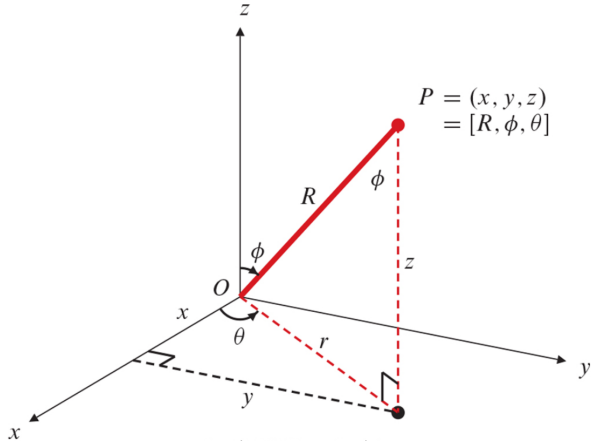
Spherical coordinates



Represented by $[R, \phi, \theta]$

$$R \geq 0, \quad \phi \in [0, \pi]$$

$$\theta \in [-\pi, \pi]$$



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Spherical coordinates (cont'd)

Definition:

$$x = R \sin \phi \cos \theta$$

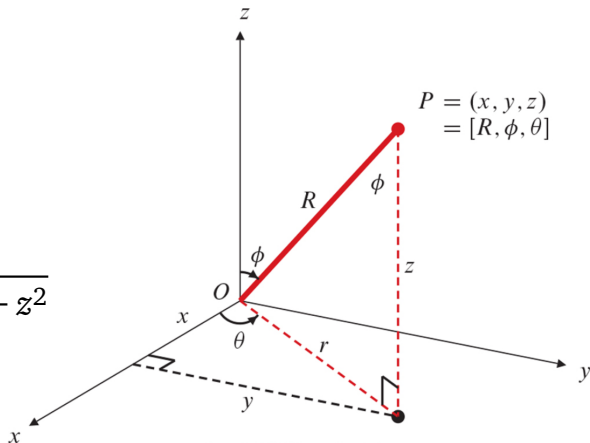
$$y = R \sin \phi \sin \theta$$

$$z = R \cos \phi$$

$$R = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$$

$$\cos \phi = \frac{z}{R}, \quad \tan \phi = \frac{r}{z}$$

$$\tan \theta = \frac{y}{x}$$



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Spherical coordinates (cont'd)

Unit vectors:

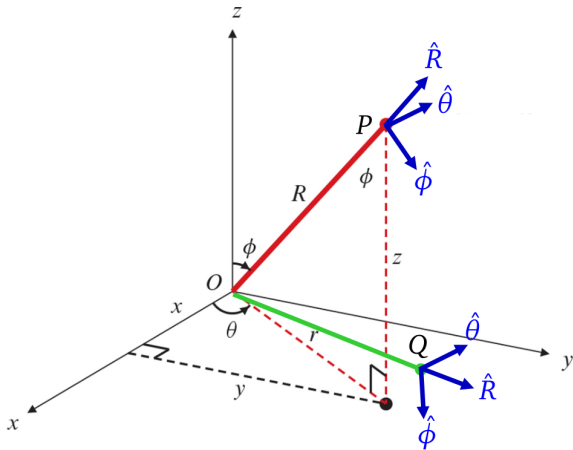
$$\hat{\mathbf{R}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{\theta}}$$

Position vector:

$$\mathbf{r}_P = R \hat{\mathbf{R}}(\phi, \theta)$$

Vector field:

$$\mathbf{A} = A_R \hat{\mathbf{R}} + A_\phi \hat{\boldsymbol{\phi}} + A_\theta \hat{\boldsymbol{\theta}}$$



Note: unit vectors relate to small changes in position vector with R , ϕ , and θ coordinates

Studio classroom problems

Studio classroom – Problem 1

Point A is located at $[r, \theta, z] = [12, \frac{3\pi}{2}, 2]$ in **cylindrical** coordinates.

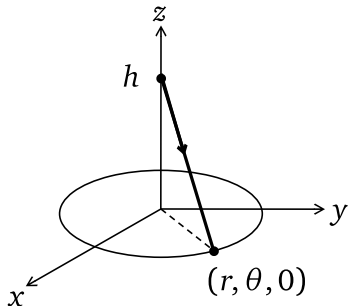
Point B is located at $[R, \phi, \theta] = [2, \frac{2\pi}{3}, \frac{\pi}{4}]$ in **spherical** coordinates.

- (a) Find the position vector $\mathbf{r}_A$ pointing from the origin to point A in Cartesian coordinates.
- (b) Find the distance between points A and B .

Studio classroom – Problem 2

The **unit** vectors $\hat{r}(\theta)$ in cylindrical coordinates and $\hat{R}(\phi, \theta)$ in spherical coordinates can be expressed in terms of Cartesian vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$ by using expressions for x , y and z .

- (a) Express $\hat{r}(\theta)$ in Cartesian components.
- (b) Express $\hat{R}(\phi, \theta)$ in Cartesian components.
- (c) Express the unit vector which points from $z = h$ on the z -axis towards the (cylindrical) point $(r, \theta, 0)$ in cylindrical coordinates.
- (d) Does the expression obtained at (c) depend on the angle θ ? If so, how?



Studio classroom – Problem 3

Consider a position vector in cylindrical coordinates $\mathbf{r} = r\hat{\mathbf{r}}(\theta)$, where both r and θ depend on time t .

- (a) Show that for the cylindrical coordinate system, it holds that

$$\frac{d\hat{\mathbf{r}}}{d\theta} = \hat{\boldsymbol{\theta}} \quad \text{and} \quad \frac{d\hat{\boldsymbol{\theta}}}{d\theta} = -\hat{\mathbf{r}}$$

- (b) Calculate the velocity $\mathbf{v} = \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt}$ and acceleration $\mathbf{a} = \dot{\mathbf{v}} = \frac{d\mathbf{v}}{dt}$.

- (c) Show that for uniform circular motion, with fixed radius a and constant angular velocity ω , we get: $\mathbf{v} = a\omega\hat{\boldsymbol{\theta}}$ and $\mathbf{a} = -a\omega^2\hat{\mathbf{r}}$.

A closer look at cylindrical coordinates

Unit vectors in plane of constant z :

$$\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}$$

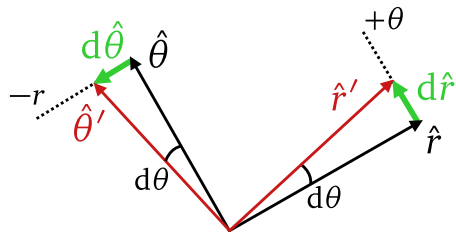
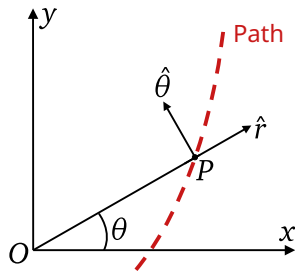
$$\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}$$

Position P on the path depends on time

- both r and θ can depend on time
- so do their unit vectors

$$\text{▸ } \frac{d\hat{r}}{d\theta} = \hat{\theta} \quad \text{and} \quad \frac{d\hat{\theta}}{d\theta} = -\hat{r}$$

$$\text{▸ } \dot{\hat{r}} = \dot{\theta} \hat{\theta} \quad \text{and} \quad \dot{\hat{\theta}} = -\dot{\theta} \hat{r}$$



Vector functions of one variable and parametrization

Studio classroom – Problem 4

The position, velocity, and acceleration of a particle moving in 3D are given by $\mathbf{r}(t)$, $\mathbf{v}(t)$, and $\mathbf{a}(t)$ as a function of time t , respectively. Furthermore, it is given that at every time t , the acceleration $\mathbf{a}$ is perpendicular to both $\mathbf{r}$ and $\mathbf{v}$.

(a) Show that the vector $\mathbf{r}(t) - t\mathbf{v}(t)$ has constant length.

Studio classroom – Problem 5

An object moves to the right along the plane curve $y = x^2$ (position in meters) with a constant *speed* of $v = 5$ m/s.

- (a) Determine the velocity and acceleration vectors of the object when it is at the position (1, 1) m.

Studio classroom – Problem 6 (if time permits)

The position of an object as a function of time t can be described by the piecewise smooth curve $\mathbf{r} = t^3\hat{\mathbf{i}} + t^2\hat{\mathbf{j}}$.

- (a) Give expressions for the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of the object as a function of time.
- (b) Calculate the distance the object has travelled (the length of the curve) between $t = -1$ and $t = 2$ s.